FEV_data_wrangling.R

User

2022-10-18

```
setwd("c:/docs/youtubeR101")
rm(list=ls())

library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2 --
## v ggplot2 3.3.6      v purrr   0.3.4
## v tibble  3.1.8      v dplyr   1.0.10
## v tidyr   1.2.1      v stringr 1.4.1
## v readr   2.1.3      v forcats 0.5.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```
fev<- haven::read_sav("FEV.sav")

dim(fev)
```

```
## [1] 654   6
```

```
glimpse(fev)
```

```
## Rows: 654
## Columns: 6
## $ Id     <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, ~
## $ Age_yr <dbl> 9, 8, 7, 9, 9, 8, 6, 6, 8, 9, 6, 8, 8, 8, 8, 7, 5, 6, 9, 9, 5, ~
## $ FEV_ml <dbl> 1.708, 1.724, 1.720, 1.558, 1.895, 2.336, 1.919, 1.415, 1.987, ~
## $ Hgt_cm <dbl> 57.0, 67.5, 54.5, 53.0, 57.0, 61.0, 58.0, 56.0, 58.5, 60.0, 53.~
## $ Sex    <dbl+lbl> 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1,~
## $ Smoke  <dbl+lbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

```
mStats::codebook(fev)
```

```
## ----------------------------------------
##        Codebook : fev
## ----------------------------------------
##   Dataset Label : -
##             Obs : 654
##            Vars : 6
```

```
##  + -- -------- + ----- ------- --- ---- +
##  | No Variable | Label    Type Obs <NA> |
##  + -- -------- + ----- ------- --- ---- +
##  |  1       Id |  <NA> numeric 654    0 |
##  |  2   Age_yr |  <NA> numeric 654    0 |
##  |  3   FEV_ml |  <NA> numeric 654    0 |
##  |  4   Hgt_cm |  <NA> numeric 654    0 |
##  |  5      Sex |  <NA>  double 654    0 |
##  |  6    Smoke |  <NA>  double 654    0 |
##  + -- -------- + ----- ------- --- ---- +
```

```
fev$Sex
```

```
## <labelled<double>[654]>
##   [1] 0 0 0 1 1 0 0 0 0 0 0 1 0 1 1 1 1 0 1 1 0 0 0 1 0 1 1 1 0 0 0 1 0 0 0 0 1
##  [38] 1 1 1 0 1 1 0 0 0 1 0 1 1 1 0 1 1 1 1 0 0 0 0 1 0 0 0 1 1 1 0 0 0 1 0 1 1
##  [75] 1 1 1 1 1 0 0 0 1 1 0 0 1 1 1 1 0 1 1 0 0 1 1 1 1 1 0 1 1 1 0 1 1 1 0 0 0
## [112] 0 1 1 0 0 0 0 0 1 1 0 0 1 1 0 1 0 0 0 1 1 1 0 1 1 1 0 1 0 1 0 1 0 1 1 1 0
## [149] 0 1 0 1 1 0 0 1 1 0 0 1 1 0 0 0 1 1 0 1 1 1 1 1 1 0 1 0 0 0 0 0 1 0 0 1 1
## [186] 1 0 0 1 1 1 0 1 1 1 0 1 0 1 1 0 0 1 0 1 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 0
## [223] 1 0 1 1 1 1 0 0 1 0 0 1 0 1 0 0 1 0 1 0 0 0 1 1 0 0 0 0 0 1 1 0 1 1 0 0 0
## [260] 1 1 0 1 1 1 1 0 0 0 0 1 1 0 1 1 0 1 1 0 1 0 1 1 0 0 0 0 1 0 1 0 0 0 1 1 1
## [297] 0 1 1 1 1 0 1 0 0 0 1 0 0 1 1 1 0 1 0 1 1 0 1 1 1 1 0 0 0 1 0 1 0 1 1 0 1
## [334] 1 0 0 0 1 0 1 1 1 1 1 0 1 0 0 1 1 1 0 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 1 1 1
## [371] 0 1 1 0 1 1 0 0 1 0 0 1 0 0 1 1 0 0 0 1 0 0 1 1 1 0 0 1 0 0 1 1 0 1 0 1 0
## [408] 0 0 0 0 1 0 0 0 1 0 0 1 1 0 1 0 1 1 1 0 1 0 0 1 0 0 0 0 1 0 0 1 0 0 1 1 1
## [445] 1 0 1 1 1 1 0 1 1 1 0 0 1 1 1 1 1 1 1 1 0 1 1 0 0 0 1 1 0 1 0 0 1 1 0 0 0
## [482] 0 0 0 1 0 0 0 1 1 1 0 0 1 1 0 0 0 0 0 0 1 0 1 0 0 0 1 1 1 0 0 1 1 1 0 1 1
## [519] 0 1 1 0 0 1 1 1 1 0 0 0 1 0 0 0 0 0 1 1 1 1 0 1 1 1 0 1 0 0 1 1 1 1 0 1 0
## [556] 1 1 1 1 0 0 0 1 1 0 0 0 1 1 0 1 1 1 0 1 0 1 1 1 0 0 0 0 1 1 0 0 1 0 0 0 0
## [593] 1 1 0 0 1 0 0 0 1 1 1 0 1 1 1 0 1 0 1 1 1 1 0 0 0 0 0 0 0 1 0 0 1 1 1 1 1
## [630] 0 0 1 0 0 0 1 1 1 0 1 1 1 0 0 1 0 1 1 1 1 1 0 0 0
##
## Labels:
##  value  label
##      0   Male
##      1 Female
```

```
fev$Sex <- as.factor(fev$Sex)

fev$Sex <- factor(fev$Sex,
                  levels = c(0,1),
                  labels = c("Male","Female"))
fev$Sex
```

```
##   [1] Male   Male   Male   Female Female Male   Male   Male   Male   Male
##  [11] Male   Female Male   Female Female Female Female Male   Female Female
##  [21] Male   Male   Male   Female Male   Female Female Female Male   Male
##  [31] Male   Female Male   Male   Male   Male   Female Female Female Female
##  [41] Male   Female Female Male   Male   Male   Female Male   Female Female
##  [51] Female Male   Female Female Female Female Male   Male   Male   Male
##  [61] Female Male   Male   Male   Female Female Female Male   Male   Male
##  [71] Female Male   Female Female Female Female Female Female Female Male
```

```
##  [81] Male   Male   Female Female Male   Male   Female Female Female Female
##  [91] Male   Female Female Male   Male   Female Female Female Female Female
## [101] Male   Female Female Female Male   Female Female Female Male   Male
## [111] Male   Male   Female Female Male   Male   Male   Male   Male   Female
## [121] Female Male   Male   Female Female Male   Female Male   Male   Male
## [131] Female Female Female Male   Female Female Female Male   Female Male
## [141] Female Male   Female Male   Female Female Female Male   Male   Female
## [151] Male   Female Female Male   Male   Female Female Male   Male   Female
## [161] Female Male   Male   Male   Female Female Male   Female Female Female
## [171] Female Female Female Male   Female Male   Male   Male   Male   Male
## [181] Female Male   Male   Female Female Female Male   Male   Female Female
## [191] Female Male   Female Female Female Male   Female Male   Female Female
## [201] Male   Male   Female Male   Female Male   Male   Male   Female Female
## [211] Male   Male   Male   Male   Male   Male   Male   Female Male   Male
## [221] Male   Male   Female Male   Female Female Female Female Male   Male
## [231] Female Male   Male   Female Male   Female Male   Male   Female Male
## [241] Female Male   Male   Male   Female Female Male   Male   Male   Male
## [251] Male   Female Female Male   Female Female Male   Male   Male   Female
## [261] Female Male   Female Female Female Female Male   Male   Male   Male
## [271] Female Female Male   Female Female Male   Female Female Male   Female
## [281] Male   Female Female Male   Male   Male   Male   Female Male   Female
## [291] Male   Male   Male   Female Female Female Male   Female Female Female
## [301] Female Male   Female Male   Male   Male   Female Male   Male   Female
## [311] Female Female Male   Female Male   Female Female Male   Female Female
## [321] Female Female Male   Male   Male   Female Male   Female Male   Female
## [331] Female Male   Female Female Male   Male   Male   Female Male   Female
## [341] Female Female Female Female Male   Female Male   Male   Female Female
## [351] Female Male   Female Female Female Female Male   Male   Male   Male
## [361] Female Female Female Female Female Male   Female Female Female Female
## [371] Male   Female Female Male   Female Female Male   Male   Female Male
## [381] Male   Female Male   Male   Female Female Male   Male   Male   Female
## [391] Male   Male   Female Female Female Male   Male   Female Male   Male
## [401] Female Female Male   Female Male   Female Male   Male   Male   Male
## [411] Male   Female Male   Male   Male   Female Male   Male   Female Female
## [421] Male   Female Male   Female Female Female Male   Female Male   Male
## [431] Female Male   Male   Male   Male   Female Male   Male   Female Male
## [441] Male   Female Female Female Female Male   Female Female Female Female
## [451] Male   Female Female Female Male   Male   Female Female Female Female
## [461] Female Female Female Female Male   Female Female Male   Male   Male
## [471] Female Female Male   Female Male   Male   Female Female Male   Male
## [481] Male   Male   Male   Male   Female Male   Male   Male   Female Female
## [491] Female Male   Male   Female Female Male   Male   Male   Male   Male
## [501] Male   Female Male   Female Male   Male   Male   Female Female Female
## [511] Male   Male   Female Female Female Male   Female Female Male   Female
## [521] Female Male   Male   Female Female Female Female Male   Male   Male
## [531] Female Male   Male   Male   Male   Male   Female Female Female Female
## [541] Male   Female Female Female Male   Female Male   Male   Female Female
## [551] Female Female Male   Female Male   Female Female Female Female Male
## [561] Male   Male   Female Female Male   Male   Male   Female Female Male
## [571] Female Female Female Male   Female Male   Female Female Female Male
## [581] Male   Male   Male   Female Female Male   Male   Female Male   Male
## [591] Male   Male   Female Female Male   Male   Female Male   Male   Male
## [601] Female Female Female Male   Female Female Female Male   Female Male
## [611] Female Female Female Female Male   Male   Male   Male   Male   Male
```

```
## [621] Female Male   Male   Female Female Female Female Female Female Male
## [631] Male   Female Male   Male   Male   Female Female Female Male   Female
## [641] Female Female Male   Male   Female Male   Female Female Female Female
## [651] Female Male   Male   Male
## Levels: Male Female
```

```
fev$Smoke
```

```
## <labelled<double>[654]>
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [112] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [149] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [186] 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [223] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [260] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [297] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0
## [334] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 1 1
## [371] 0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0
## [408] 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 0 0
## [445] 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0
## [482] 0 1 1 0 0 0 1 0 0 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1
## [519] 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
## [556] 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
## [593] 0 1 1 0 0 0 0 1 1 1 0 1 0 0 1 0 0 1 1 0 0 0 0 1 1 1 0 0 1 1 1 0 0 0 0 0 1
## [630] 0 0 0 1 1 1 0 1 1 0 1 0 0 0 1 1 1 0 0 1 1 1 0 1 0
##
## Labels:
##  value       label
##      0 Not exposed
##      1     Exposed
```

```
fev$Smoke <- as.factor(fev$Smoke)

fev$Smoke <- factor(fev$Smoke,
                    levels = c(0,1),
                    labels = c("Not Exposed","Exposed"))

fev$Smoke
```

```
##   [1] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##   [7] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [13] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [19] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [25] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [31] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [37] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [43] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [49] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [55] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [61] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [67] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
```

```
##  [73] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [79] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [85] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [91] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
##  [97] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [103] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [109] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [115] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [121] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [127] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [133] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [139] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [145] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [151] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [157] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [163] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [169] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [175] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [181] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [187] Not Exposed Not Exposed Not Exposed Not Exposed Exposed     Not Exposed
## [193] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [199] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [205] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [211] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [217] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [223] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [229] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [235] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [241] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [247] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [253] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [259] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [265] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [271] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [277] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [283] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [289] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [295] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [301] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [307] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [313] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [319] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [325] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [331] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Not Exposed
## [337] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [343] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [349] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [355] Not Exposed Not Exposed Not Exposed Exposed     Not Exposed Not Exposed
## [361] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Exposed
## [367] Not Exposed Not Exposed Exposed     Exposed     Not Exposed Exposed
## [373] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [379] Not Exposed Not Exposed Exposed     Not Exposed Not Exposed Exposed
## [385] Not Exposed Not Exposed Not Exposed Exposed     Not Exposed Not Exposed
## [391] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
```

```
## [397] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [403] Exposed     Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [409] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Exposed
## [415] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [421] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Not Exposed
## [427] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [433] Not Exposed Not Exposed Exposed     Not Exposed Not Exposed Not Exposed
## [439] Exposed     Not Exposed Exposed     Not Exposed Not Exposed Not Exposed
## [445] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Not Exposed
## [451] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Exposed
## [457] Not Exposed Not Exposed Not Exposed Not Exposed Exposed     Not Exposed
## [463] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [469] Not Exposed Not Exposed Not Exposed Exposed     Not Exposed Not Exposed
## [475] Not Exposed Not Exposed Not Exposed Not Exposed Exposed     Not Exposed
## [481] Not Exposed Not Exposed Exposed     Exposed     Not Exposed Not Exposed
## [487] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Not Exposed
## [493] Not Exposed Exposed     Not Exposed Exposed     Not Exposed Exposed
## [499] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [505] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Not Exposed
## [511] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [517] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Exposed
## [523] Exposed     Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [529] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [535] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [541] Exposed     Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [547] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [553] Not Exposed Not Exposed Not Exposed Exposed     Not Exposed Not Exposed
## [559] Exposed     Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [565] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [571] Not Exposed Not Exposed Not Exposed Exposed     Not Exposed Not Exposed
## [577] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [583] Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed Not Exposed
## [589] Not Exposed Exposed     Not Exposed Not Exposed Not Exposed Exposed
## [595] Exposed     Not Exposed Not Exposed Not Exposed Not Exposed Exposed
## [601] Exposed     Exposed     Not Exposed Exposed     Not Exposed Not Exposed
## [607] Exposed     Not Exposed Not Exposed Exposed     Exposed     Not Exposed
## [613] Not Exposed Not Exposed Not Exposed Exposed     Exposed     Exposed
## [619] Not Exposed Not Exposed Exposed     Exposed     Exposed     Not Exposed
## [625] Not Exposed Not Exposed Not Exposed Not Exposed Exposed     Not Exposed
## [631] Not Exposed Not Exposed Exposed     Exposed     Exposed     Not Exposed
## [637] Exposed     Exposed     Not Exposed Exposed     Not Exposed Not Exposed
## [643] Not Exposed Exposed     Exposed     Exposed     Not Exposed Not Exposed
## [649] Exposed     Exposed     Exposed     Not Exposed Exposed     Not Exposed
## Levels: Not Exposed Exposed
```

```
is.na(fev$Id)
```

```
##   [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [37] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [49] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
##  [85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
##  [97] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [109] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [121] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [133] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [145] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [157] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [169] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [181] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [193] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [205] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [217] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [229] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [241] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [253] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [265] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [277] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [301] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [313] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [325] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [337] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [349] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [361] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [373] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [385] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [397] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [409] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [421] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [433] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [445] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [457] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [469] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [481] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [493] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [505] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [517] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [529] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [541] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [553] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [565] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [577] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [589] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [601] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [613] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [625] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [637] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [649] FALSE FALSE FALSE FALSE FALSE FALSE
```

```
view(is.na.data.frame(fev))

fivenum(fev$Age_yr)
```

```
## [1]  3  8 10 12 19
```

```r
mean(fev$Age_yr, na.rm = T)
```

```
## [1] 9.931193
```

```r
mStats::summ(fev, Age_yr)
```

```
##   Summary
##  + -------- + --- ---- ---- ------- --- ---- ------- +
##  | Variable | Obs <NA> Mean Std.Dev Min  Max Normal. |
##  + -------- + --- ---- ---- ------- --- ---- ------- +
##  |   Age_yr | 654    0  9.9     3.0 3.0 19.0  0.0000 |
##  + -------- + --- ---- ---- ------- --- ---- ------- +
```

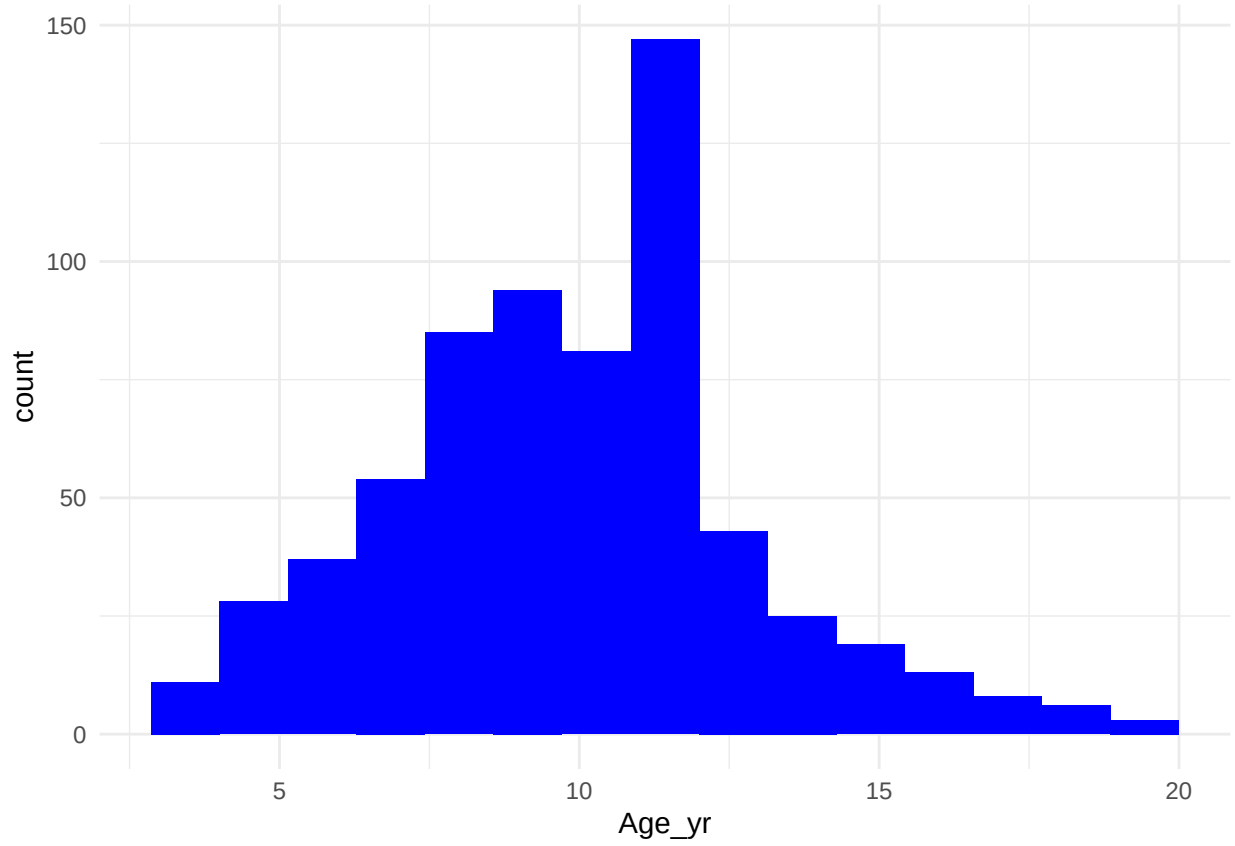
```r
raster::modal(fev$Age_yr,na.rm=TRUE)
```

```
## [1] 9
```

```r
t.test(fev$Age_yr) # 95%CI for mean
```

```
##
##  One Sample t-test
##
## data:  fev$Age_yr
## t = 85.978, df = 653, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##   9.704381 10.158005
## sample estimates:
## mean of x
##  9.931193
```

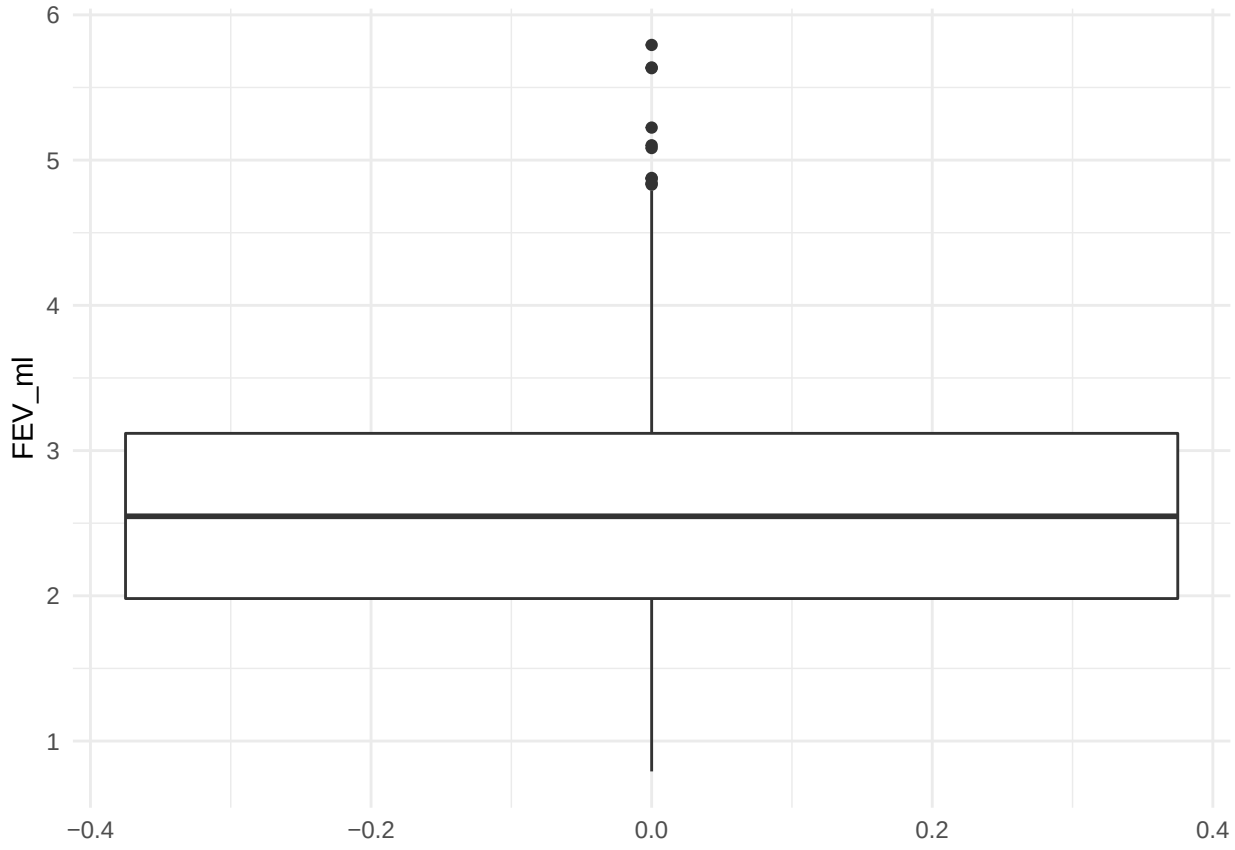
```r
fev %>%
  ggplot(aes(Age_yr))+
  geom_histogram(bins = 15, fill= "blue")+
 theme_minimal()
```




```
fivenum(fev$FEV_ml)
```

```
## [1] 0.7910 1.9790 2.5475 3.1200 5.7930
```

```
fev %>%
  ggplot(aes(FEV_ml))+
  geom_boxplot()+
  coord_flip()+
  theme_minimal()
```




```
mStats::tab(fev, Sex,Smoke)
```

```
##   One-way Tabulation
##  + -------- ----------- + ----- ------- ----- +
##  | Variable    Category | Freq. Percent  Cum. |
##  + -------- ----------- + ----- ------- ----- +
##  |      Sex        Male |   318    48.6  48.6 |
##  |               Female |   336    51.4 100.0 |
##  |          ----------- | ----- ------- ----- |
##  |                Total |   654     100   100 |
##  | -------- ----------- | ----- ------- ----- |
##  |    Smoke Not Exposed |   589    90.1  90.1 |
##  |              Exposed |    65     9.9 100.0 |
##  |          ----------- | ----- ------- ----- |
##  |                Total |   654     100   100 |
##  + -------- ----------- + ----- ------- ----- +
```

```
mStats::tab(fev, Sex,by= Smoke)
```

```
##   Tabulation by 'Smoke'
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  | Variable Category | Not Exposed r(%) Exposed r(%) Total r(%) | Chi-squared
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  |      Sex     Male |         279 87.7      39 12.3   318  100 |       0.053
##  |            Female |         310 92.3      26  7.7   336  100 |
```

```
##  |          -------- | ----------- ---- ------- ---- ----- ---- | -----------
##  |             Total |         589 90.1      65  9.9   654  100 |
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  ------ +
##  Exact. |
##  ------ +
##   0.067 |
##         |
##  ------ |
##         |
##  ------ +
```

```
mStats::summ(fev, FEV_ml,Hgt_cm)
```

```
##   Summary
##  + -------- + --- ---- ---- ------- ---- ---- ------- +
##  | Variable | Obs <NA> Mean Std.Dev  Min  Max Normal. |
##  + -------- + --- ---- ---- ------- ---- ---- ------- +
##  |   FEV_ml | 654    0  2.6     0.9  0.8  5.8  0.0000 |
##  |   Hgt_cm | 654    0 61.1     5.7 46.0 74.0  0.0001 |
##  + -------- + --- ---- ---- ------- ---- ---- ------- +
```

```
mStats::summ(fev, FEV_ml, by= Sex)
```

```
##   Summary by 'Sex'
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
##  | Variable  Level | Obs <NA> Mean Std.Dev Normal. t-test Wilcoxon |
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
##  |   FEV_ml Female | 336    0  2.8     1.0  0.0000 0.0000   0.0002 |
##  |            Male | 318    0  2.5     0.6  0.0025                  |
##  |          ------ | --- ---- ---- ------- ------- ------ -------- |
##  |           Total | 654    0  2.6     0.9  0.0000                  |
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
```

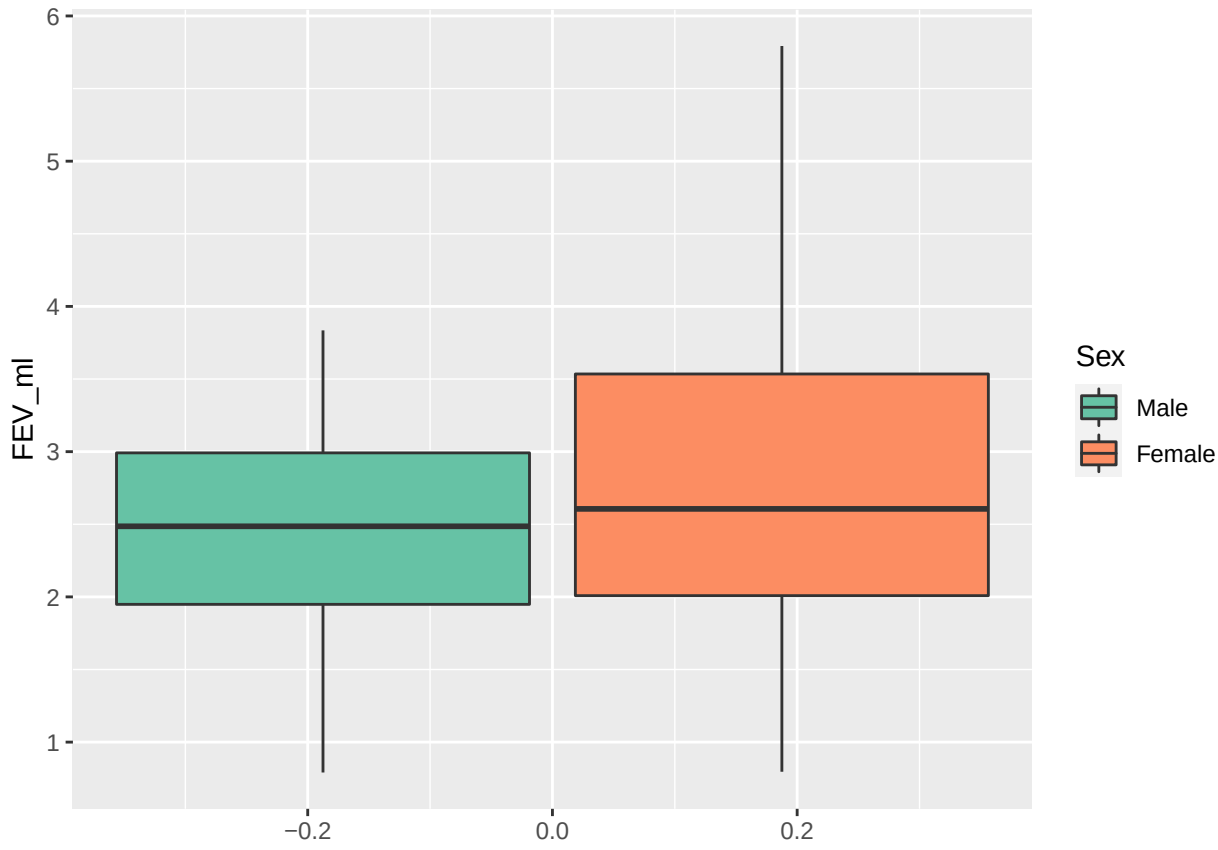
```
mStats::summ(fev, FEV_ml, by= Smoke)
```

```
##   Summary by 'Smoke'
##  + -------- ----------- + --- ---- ---- ------- ------- ------ -------- +
##  | Variable       Level | Obs <NA> Mean Std.Dev Normal. t-test Wilcoxon |
##  + -------- ----------- + --- ---- ---- ------- ------- ------ -------- +
##  |   FEV_ml     Exposed |  65    0  3.3     0.7  0.1427 0.0000   0.0000 |
##  |          Not Exposed | 589    0  2.6     0.9  0.0000                 |
##  |          ----------- | --- ---- ---- ------- ------- ------ -------- |
##  |                Total | 654    0  2.6     0.9  0.0000                 |
##  + -------- ----------- + --- ---- ---- ------- ------- ------ -------- +
```

```
##############################################################################

#Q1
# mean diff on sex
```

```
fev %>%
  ggplot(aes(FEV_ml, fill= Sex))+
  geom_boxplot()+
  coord_flip()+
  scale_fill_brewer(palette = "Set2")
```




```
#ttest

mStats::summ(fev, FEV_ml, by= Sex)
```

```
##   Summary by 'Sex'
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
##  | Variable  Level | Obs <NA> Mean Std.Dev Normal. t-test Wilcoxon |
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
##  |   FEV_ml Female | 336    0  2.8     1.0  0.0000 0.0000   0.0002 |
##  |            Male | 318    0  2.5     0.6  0.0025                  |
##  |          ------ | --- ---- ---- ------- ------- ------ -------- |
##  |           Total | 654    0  2.6     0.9  0.0000                  |
##  + -------- ------ + --- ---- ---- ------- ------- ------ -------- +
```

```
t.test(FEV_ml~Sex,data=fev,alternative= "greater",var.equal=TRUE)
```

```
##
##  Two Sample t-test
```

```
##
## data:  FEV_ml by Sex
## t = -5.4412, df = 652, p-value = 1
## alternative hypothesis: true difference in means between group Male and group Female is greater than
## 95 percent confidence interval:
##  -0.4706442        Inf
## sample estimates:
##   mean in group Male mean in group Female
##             2.451170             2.812446
```

```
#Q2

fev <- fev %>%
  mutate(Age_gp= case_when(Age_yr<=5 ~ "<5", 5<Age_yr & Age_yr<13 ~"6-12", Age_yr>12 ~ ">12"))

fev$Age_gp<- as.factor(fev$Age_gp)

fev$Age_gp
```

```
##   [1] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
##  [16] 6-12 <5   6-12 6-12 6-12 <5   <5   <5   6-12 6-12 <5   6-12 <5   6-12 6-12
##  [31] <5   6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5
##  [46] 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
##  [61] 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12
##  [76] 6-12 <5   <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
##  [91] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
## [106] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12
## [121] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [136] 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [151] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12
## [166] 6-12 6-12 6-12 6-12 <5   6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 <5   6-12
## [181] 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [196] 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12
## [211] 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5   6-12 6-12 6-12 <5   6-12 6-12 6-12
## [226] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [241] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [256] 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5
## [271] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5   6-12 6-12 6-12 6-12 6-12
## [286] <5   6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
## [301] 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12
## [316] 6-12 6-12 6-12 >12  >12  >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12
## [331] >12  >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [346] 6-12 6-12 6-12 >12  >12  6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  >12  6-12
## [361] >12  6-12 6-12 >12  6-12 >12  6-12 >12  6-12 >12  6-12 >12  6-12 6-12 6-12
## [376] 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 >12
## [391] >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  >12  6-12 6-12 6-12
## [406] 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12
## [421] 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 >12  6-12 6-12 6-12 >12  6-12 >12
## [436] 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 >12
## [451] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12
## [466] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 >12  6-12
## [481] 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 >12  >12  6-12 6-12 6-12 >12
## [496] 6-12 6-12 >12  6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12
## [511] 6-12 6-12 6-12 >12  6-12 >12  >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
```

```
## [526] 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12
## [541] >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12
## [556] >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12
## [571] 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12
## [586] 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 >12  6-12 6-12 >12  6-12 6-12
## [601] >12  >12  6-12 6-12 6-12 >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [616] >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [631] >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [646] >12  >12  >12  >12  >12  >12  >12  >12  >12
## Levels: <5 >12 6-12
```

```
fev$Age_gp <- factor(fev$Age_gp,
                     levels =c("<5","6-12",">12"))

fev$Age_gp
```

```
##   [1] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
##  [16] 6-12 <5   6-12 6-12 6-12 <5   <5   <5   6-12 6-12 <5   6-12 <5   6-12 6-12
##  [31] <5   6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5
##  [46] 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
##  [61] 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12
##  [76] 6-12 <5   <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
##  [91] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
## [106] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12
## [121] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [136] 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [151] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12
## [166] 6-12 6-12 6-12 6-12 <5   6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 <5   6-12
## [181] 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [196] 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12
## [211] 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5   6-12 6-12 6-12 <5   6-12 6-12 6-12
## [226] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [241] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [256] 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5
## [271] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 <5   6-12 6-12 6-12 6-12 6-12
## [286] <5   6-12 <5   6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12
## [301] 6-12 6-12 6-12 6-12 6-12 6-12 <5   6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12
## [316] 6-12 6-12 6-12 >12  >12  >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12
## [331] >12  >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [346] 6-12 6-12 6-12 >12  >12  6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  >12  6-12
## [361] >12  6-12 6-12 >12  6-12 >12  6-12 >12  6-12 >12  6-12 >12  6-12 6-12 6-12
## [376] 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 >12
## [391] >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  >12  6-12 6-12 6-12
## [406] 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12
## [421] 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 >12  6-12 6-12 6-12 >12  6-12 >12
## [436] 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 >12
## [451] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12
## [466] 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 6-12 >12  6-12
## [481] 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 >12  >12  6-12 6-12 6-12 >12
## [496] 6-12 6-12 >12  6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12
## [511] 6-12 6-12 6-12 >12  6-12 >12  >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12
## [526] 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12
## [541] >12  6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12 6-12 >12
## [556] >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12
```

```
## [571] 6-12 >12  6-12 6-12 6-12 6-12 6-12 6-12 6-12 6-12 >12  6-12 6-12 6-12
## [586] 6-12 6-12 6-12 >12  6-12 6-12 6-12 >12  6-12 >12  6-12 6-12 >12  6-12 6-12
## [601] >12  >12  6-12 6-12 6-12 >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [616] >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [631] >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12  >12
## [646] >12  >12  >12  >12  >12  >12  >12  >12  >12
## Levels: <5 6-12 >12
```

```
mStats::summ(fev, FEV_ml, by= Age_gp)
```

```
##   Summary by 'Age_gp'
##  + -------- ----- + --- ---- ---- ------- ------- ------ -------- +
##  | Variable Level | Obs <NA> Mean Std.Dev Normal.  ANOVA K-Wallis |
##  + -------- ----- + --- ---- ---- ------- ------- ------ -------- +
##  |   FEV_ml    <5 |  39    0  1.5     0.3  0.6561 0.0000   0.0000 |
##  |            >12 | 117    0  3.6     0.8  0.0478                 |
##  |           6-12 | 498    0  2.5     0.7  0.0000                 |
##  |          ----- | --- ---- ---- ------- ------- ------ -------- |
##  |          Total | 654    0  2.6     0.9  0.0000                 |
##  + -------- ----- + --- ---- ---- ------- ------- ------ -------- +
```

```
myanova1<-aov(FEV_ml~Age_gp,data=fev)
summary(myanova1)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Age_gp        2  170.4   85.19     173 <2e-16 ***
## Residuals   651  320.5    0.49
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(myanova1)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = FEV_ml ~ Age_gp, data = fev)
##
## $Age_gp
##              diff       lwr      upr p adj
## 6-12-<5  1.029419 0.7553280 1.303509     0
## >12-<5   2.127043 1.8222596 2.431826     0
## >12-6-12 1.097624 0.9282747 1.266974     0
```

```
#Q3

fev <- fev %>%
  mutate(FEV_gp= case_when(FEV_ml<=2 ~ "1", FEV_ml>2 ~ "2"))

fev$FEV_gp<- as.factor(fev$FEV_gp)

fev$FEV_gp <- factor(fev$FEV_gp,
```

```
                    levels =c(1,2),
                    labels = c("Impaired","Normal"))

fev$FEV_gp
```

```
##   [1] Impaired Impaired Impaired Impaired Impaired Normal   Impaired Impaired
##   [9] Impaired Impaired Impaired Impaired Normal   Normal   Normal   Impaired
##  [17] Impaired Impaired Normal   Normal   Impaired Impaired Impaired Normal
##  [25] Normal   Impaired Normal   Impaired Normal   Normal   Impaired Normal
##  [33] Normal   Normal   Impaired Normal   Normal   Normal   Impaired Normal
##  [41] Normal   Normal   Impaired Impaired Impaired Normal   Impaired Impaired
##  [49] Impaired Normal   Normal   Normal   Normal   Impaired Impaired Normal
##  [57] Normal   Normal   Impaired Impaired Normal   Normal   Impaired Impaired
##  [65] Impaired Impaired Normal   Impaired Impaired Normal   Impaired Normal
##  [73] Normal   Normal   Normal   Impaired Impaired Impaired Normal   Impaired
##  [81] Normal   Impaired Normal   Impaired Impaired Normal   Impaired Normal
##  [89] Impaired Normal   Normal   Normal   Impaired Normal   Normal   Normal
##  [97] Impaired Normal   Impaired Impaired Impaired Impaired Normal   Impaired
## [105] Normal   Impaired Normal   Impaired Impaired Impaired Normal   Impaired
## [113] Normal   Normal   Impaired Normal   Impaired Impaired Impaired Normal
## [121] Normal   Impaired Impaired Impaired Normal   Impaired Impaired Normal
## [129] Impaired Normal   Normal   Impaired Impaired Impaired Normal   Impaired
## [137] Impaired Normal   Impaired Impaired Impaired Normal   Impaired Normal
## [145] Normal   Normal   Impaired Normal   Normal   Normal   Normal   Normal
## [153] Normal   Normal   Impaired Impaired Impaired Normal   Normal   Normal
## [161] Normal   Impaired Impaired Impaired Normal   Impaired Normal   Impaired
## [169] Normal   Impaired Impaired Normal   Impaired Normal   Normal   Impaired
## [177] Normal   Normal   Impaired Impaired Impaired Normal   Normal   Normal
## [185] Impaired Impaired Impaired Normal   Normal   Impaired Impaired Impaired
## [193] Normal   Impaired Impaired Normal   Normal   Impaired Normal   Normal
## [201] Normal   Impaired Normal   Normal   Impaired Normal   Impaired Normal
## [209] Impaired Normal   Normal   Normal   Normal   Impaired Normal   Impaired
## [217] Normal   Impaired Normal   Impaired Impaired Impaired Normal   Impaired
## [225] Impaired Normal   Impaired Impaired Impaired Impaired Normal   Normal
## [233] Impaired Impaired Normal   Normal   Normal   Impaired Normal   Impaired
## [241] Impaired Normal   Normal   Impaired Normal   Normal   Impaired Normal
## [249] Impaired Impaired Impaired Normal   Normal   Normal   Normal   Normal
## [257] Normal   Impaired Impaired Normal   Normal   Impaired Normal   Impaired
## [265] Normal   Normal   Impaired Impaired Impaired Impaired Normal   Normal
## [273] Normal   Normal   Impaired Impaired Normal   Impaired Normal   Impaired
## [281] Normal   Impaired Normal   Normal   Normal   Impaired Normal   Impaired
## [289] Impaired Normal   Normal   Normal   Impaired Impaired Normal   Impaired
## [297] Impaired Impaired Impaired Impaired Impaired Normal   Impaired Impaired
## [305] Normal   Normal   Normal   Normal   Impaired Normal   Normal   Normal
## [313] Normal   Normal   Normal   Impaired Normal   Normal   Normal   Normal
## [321] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [329] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [337] Normal   Normal   Normal   Normal   Impaired Normal   Normal   Normal
## [345] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [353] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [361] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [369] Impaired Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [377] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
```

```
## [385] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [393] Normal   Normal   Normal   Normal   Impaired Normal   Normal   Normal
## [401] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [409] Normal   Normal   Normal   Normal   Normal   Normal   Impaired Normal
## [417] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [425] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [433] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [441] Normal   Normal   Normal   Impaired Impaired Normal   Normal   Normal
## [449] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [457] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [465] Normal   Normal   Normal   Normal   Normal   Normal   Impaired Normal
## [473] Normal   Normal   Normal   Normal   Normal   Impaired Normal   Normal
## [481] Normal   Normal   Normal   Normal   Impaired Normal   Normal   Normal
## [489] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [497] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [505] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [513] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [521] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [529] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [537] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [545] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [553] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [561] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [569] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [577] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [585] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [593] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [601] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [609] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [617] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [625] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [633] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [641] Normal   Normal   Normal   Normal   Normal   Normal   Normal   Normal
## [649] Normal   Normal   Normal   Normal   Normal   Normal
## Levels: Impaired Normal
```

```
mStats::tab(fev, FEV_gp, by= Smoke)
```

```
##   Tabulation by 'Smoke'
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  | Variable Category | Not Exposed r(%) Exposed r(%) Total r(%) | Chi-squared
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  |   FEV_gp Impaired |         165 98.8       2  1.2   167  100 |       0.000
##  |           Normal |         424 87.1      63 12.9   487  100 |
##  |          -------- | ----------- ---- ------- ---- ----- ---- | -----------
##  |            Total |         589 90.1      65  9.9   654  100 |
##  + -------- -------- + ----------- ---- ------- ---- ----- ---- + -----------
##  ------ +
##  Exact. |
##  ------ +
##   0.000 |
##         |
##  ------ |
##         |
```

```
##  ------ +
```

```
#Q4

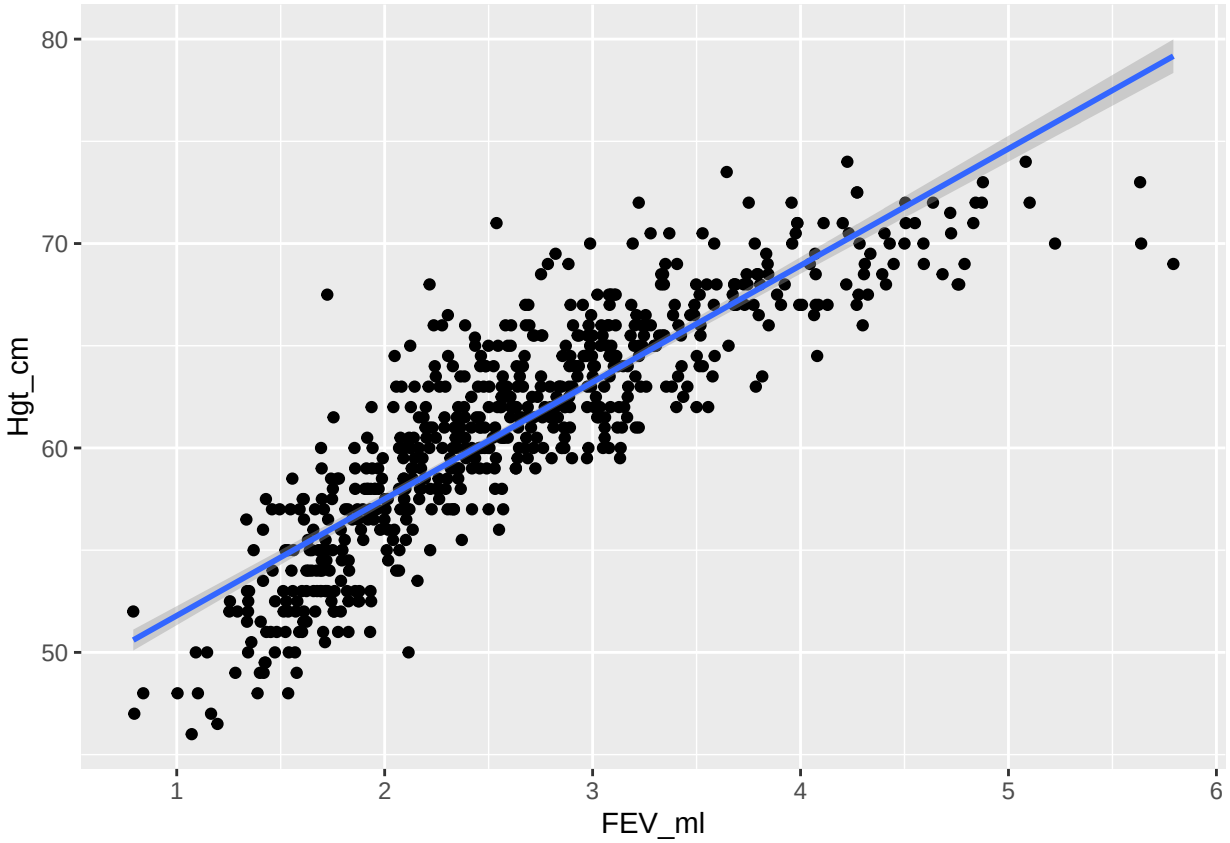
mStats::tab(fev, FEV_gp, by= Age_gp)
```

```
##   Tabulation by 'Age_gp'
##  + -------- -------- + -- ---- ---- ---- --- ---- ----- ---- + -----------
##  | Variable Category | <5 r(%) 6-12 r(%) >12 r(%) Total r(%) | Chi-squared
##  + -------- -------- + -- ---- ---- ---- --- ---- ----- ---- + -----------
##  |   FEV_gp Impaired | 37 22.2  130 77.8   0    0   167  100 |       0.000
##  |            Normal |  2  0.4  368 75.6 117   24   487  100 |
##  |          -------- | -- ---- ---- ---- --- ---- ----- ---- | -----------
##  |             Total | 39    6  498 76.1 117 17.9   654  100 |
##  + -------- -------- + -- ---- ---- ---- --- ---- ----- ---- + -----------
##  ------ +
##  Exact. |
##  ------ +
##   0.000 |
##         |
##  ------ |
##         |
##  ------ +
```

```
#Q5

fev %>%
  ggplot(aes(FEV_ml,Hgt_cm))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula 'y ~ x'
```




```
# Linear correlation
cor.test(fev$FEV_ml,fev$Hgt_cm,method="pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  fev$FEV_ml and fev$Hgt_cm
## t = 44.662, df = 652, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8479023 0.8858426
## sample estimates:
##      cor
## 0.868135
```

```
#Q6

est1 <- lm(FEV_ml~Age_yr,data=fev) ; est1
```

```
##
## Call:
## lm(formula = FEV_ml ~ Age_yr, data = fev)
##
## Coefficients:
## (Intercept)       Age_yr
```

```
##      0.4316       0.2220
```

```
#Q7


cd4<- haven::read_sav("CD4_paired.sav")

dim(cd4)
```

```
## [1] 7 3
```

```
is.na(cd4)
```

```
##         ID Before_CD4 After_CD4
## [1,] FALSE      FALSE     FALSE
## [2,] FALSE      FALSE     FALSE
## [3,] FALSE      FALSE     FALSE
## [4,] FALSE      FALSE     FALSE
## [5,] FALSE      FALSE     FALSE
## [6,] FALSE      FALSE     FALSE
## [7,] FALSE      FALSE     FALSE
```

```
b4<- cd4$Before_CD4
af<- cd4$After_CD4

t.test(af,b4,paired=TRUE)
```

```
##
##  Paired t-test
##
## data:  af and b4
## t = 4.46, df = 6, p-value = 0.004283
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##   65.19018 223.66696
## sample estimates:
## mean difference
##        144.4286
```

```
wilcox.test(af,b4,paired=TRUE)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  af and b4
## V = 28, p-value = 0.01563
## alternative hypothesis: true location shift is not equal to 0
```